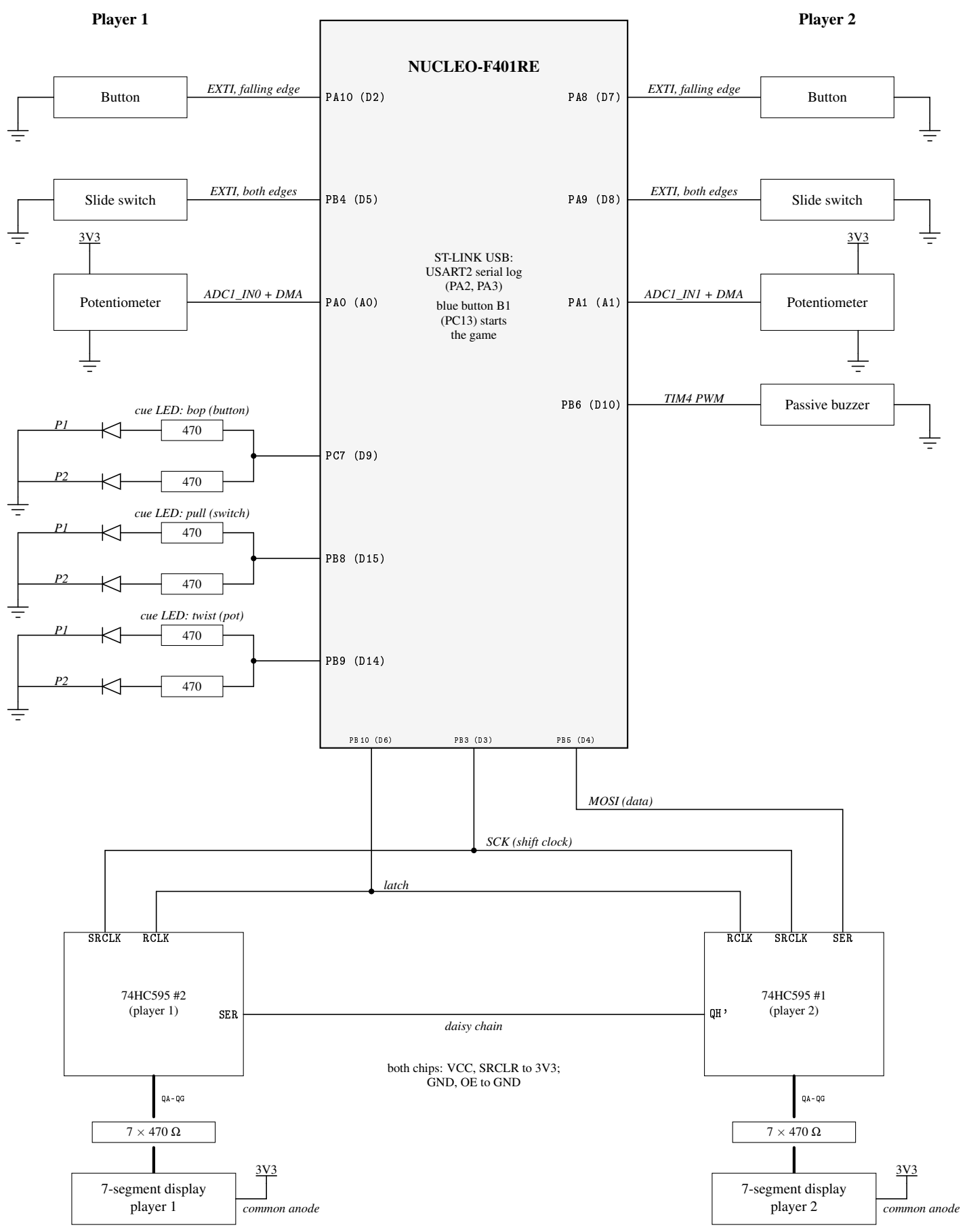

Player 1
Player 2
NUCLEO-F401RE
Button
EXTI, falling edge
PA10 (D2)
PA8 (D7)
EXTI, falling edge
Button
Slide switch
EXTI, both edges
PB4 (D5)
PA9 (D8)
EXTI, both edges
Slide switch
3V3
3V3
Potentiometer
ADC1_IN0 + DMA
PA0 (A0)
PA1 (A1)
ADC1_IN1 + DMA
Potentiometer
ST-LINK USB:
USART2 serial log
(PA2, PA3)
blue button B1
(PC13) starts
the game
PB6 (D10)
TIM4 PWM
Passive buzzer
cue LED: bop (button)
P1
470
P2
470
PC7 (D9)
cue LED: pull (switch)
P1
470
P2
470
PB8 (D15)
cue LED: twist (pot)
P1
470
P2
470
PB9 (D14)
PB10 (D6)
PB3 (D3)
PB5 (D4)
MOSI (data)
SCK (shift clock)
latch
SRCLK
RCLK
74HC595 #2
(player 1)
SER
RCLK
SRCLK
SER
74HC595 #1
(player 2)
QH'
daisy chain
both chips: VCC, SRCLR to 3V3;
GND, OE to GND
QA-QG
7 × 470 Ω
3V3
7-segment display
player 1
common anode
QA-QG
7 × 470 Ω
3V3
7-segment display
player 2
common anode